

Motion Cues for Robust Continuous Sign Language Recognition: Baselines and Diagnostics

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Abstract—Continuous Sign Language Recognition (CSLR) is sensitive to how motion unfolds over time: short stroke-like segments often carry linguistic content, while transitional movements connect signs and can introduce boundary ambiguity. Standard CSLR systems typically learn these temporal regimes implicitly using a CNN backbone and a temporal module trained with Connectionist Temporal Classification (CTC), which can make decoding prone to deletion/insertion errors near boundaries. In this paper, we present an empirical study of CTC based CSLR on PHOENIX-2014, reporting Word Error Rate (WER) together with deletion/insertion breakdowns for standard backbone-temporal baselines. We additionally introduce a lightweight motion saliency diagnostic based on normalized frame differences to highlight candidate boundary-difficult intervals and support discussion/visual inspection. Finally, we describe a standard self distillation setup that stabilizes training without requiring extra annotations. The overall goal is to characterize boundary-related error patterns and motivate boundary-aware modeling directions for improving CSLR alignment robustness.

Index Terms—continuous sign language recognition, CTC, temporal modeling, motion cues, error analysis, knowledge distillation

I. INTRODUCTION

Continuous sign language recognition (CSLR) turns long sign videos into gloss sequences and helps reduce the communication gap between deaf and hearing communities [1]. Recent systems have improved quickly with CNN + temporal models trained using CTC [2] [3] [4]. Many of them try to use motion better: CorrNet matches neighboring frames to find displacement [5]; SEN picks key frames using motion cues [6]; SignGraph models relations across and within frames with graphs [8]; and OLMD splits motion into horizontal and vertical parts [7].

A practical challenge in CSLR is that not all motion is equally informative. In continuous signing, stroke-like segments often carry lexical evidence, while transitional movements mainly connect signs. Around these boundaries, visual evidence can be ambiguous due to co-articulation, rapid hand motion, and signer-dependent variability. When such regimes are treated uniformly, the temporal decoder can receive noisy or uncertain evidence, which may increase CTC

deletion and insertion errors.

Motivated by this stroke-transition perspective, this paper focuses on documenting practical error patterns in CTC-based CSLR using standard backbone+temporal baselines, reporting WER together with deletion/insertion breakdowns. We also include a lightweight motion-saliency visualization based on normalized frame differences as an illustrative tool for discussing where pixel-level motion is strong in CSLR videos. Finally, we describe a standard self-distillation (KD) setup that uses intermediate logits as an auxiliary training signal, without requiring additional annotations.

The key contributions of this paper are as follows:

- A stroke-transition perspective on CSLR motion that motivates boundary-aware discussion in CTC-based recognition.
- An empirical report on PHOENIX-2014 with WER and del/ins breakdowns for representative CSLR baselines.
- A lightweight motion-saliency visualization (normalized frame differences) and a standard self-distillation setup described in a reproducible manner.

II. RELATED WORK

A. Continuous Sign Language Recognition

Modern CSLR typically combines a visual backbone (2D/3D CNN) with a temporal module (1D CNN or BiLSTM) [13] and learns alignments using CTC, which removes the need for frame-level gloss labels while enabling end-to-end training [9]. Recent systems keep this recipe but vary the spatial modeling (e.g., relation/graph reasoning) and temporal heads; e.g., SignGraph organizes intra-/inter-frame relations while still decoding with a CTC-style head [8].

B. Frame-Level Difference as a Motion Cue

Adding an explicit motion signal can sharpen temporal evidence. A simple option is frame differencing (or optical flow) alongside RGB. CorrNet computes correlations between adjacent frames to better capture motion displacement, improving alignment and recognition on continuous signing [5]. Segmentation-oriented work also shows that lightweight per-frame differences/flow can help surface boundaries and motion

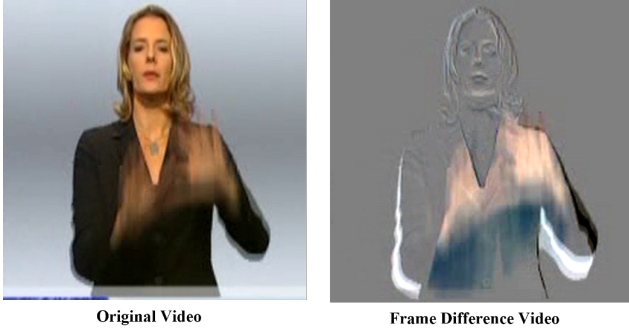


Fig. 1. Original frame (left) and normalized frame-difference map (right), illustrating pixel-level motion saliency in a CSLR video.

saliency, particularly in shallower encoders or when domain shift is present [10]. Classic two-stream vision results further support the value of an explicit motion channel [11].

C. Transition-Aware Modeling

Beyond “more motion,” continuous signing benefits from explicit modeling of transitions (low-content movements between strokes). Linguistically motivated segmentation argues for boundary-aware labeling (BIO rather than IO) and ties motion/prosodic cues to phrase boundaries, showing cleaner segment recovery across frame rates and domains [10]. This motivates our focus on boundary ambiguity in CTC-based CSLR and the use of motion-oriented visualizations when discussing transition regions.

III. BASELINE AND TRAINING SETUP

Let a sign video be $X = \{x_t\}_{t=1}^T$, $x_t \in \mathbb{R}^{H \times W \times 3}$. Our system follows a standard CSLR recipe: an appearance encoder extracts per-frame features, a temporal head aggregates context, and a CTC classifier produces per-timestep logits for decoding. In addition, we optionally use an auxiliary CTC head on an intermediate representation to provide a self-distillation (KD) signal during training (Fig. 2). For qualitative illustration, we also visualize motion saliency using normalized frame differences (Fig. 1). These components require no additional annotations.

A. Appearance Encoder (ResNet)

Each frame is encoded by a 2D CNN f_θ to features $\alpha_t \in \mathbb{R}^d$:

$$\alpha_t = f_\theta(x_t), \quad A = [\alpha_1, \dots, \alpha_T] \in \mathbb{R}^{T \times d} \quad (1)$$

We use a plain ResNet-34/18 backbone and drop the classification head [12].

B. Temporal Head (1D CNN $\rightarrow$ BiLSTM)

A temporal 1D CNN g_ψ refines per-frame features,

$$u_{1:T'} = g_\psi(A) \in \mathbb{R}^{T' \times d}, \quad (2)$$

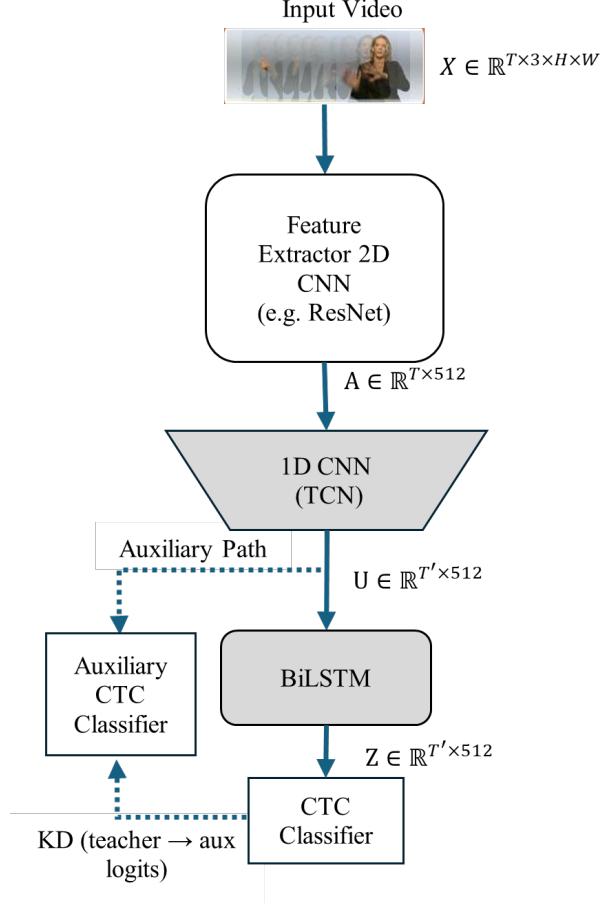


Fig. 2. Overall pipeline. A standard CSLR backbone (2DCNN $\rightarrow$ temporal module $\rightarrow$ BiLSTM) is trained with a CTC classifier. An auxiliary CTC head attached to an intermediate representation is trained via self-distillation from the final CTC logits.

followed by a BiLSTM h_ϕ to obtain context-aware representations,

$$Z = h_\phi(u_{1:T'}) = [z_1, \dots, z_{T'}] \in \mathbb{R}^{T' \times d}. \quad (3)$$

This “appearance $\rightarrow$ 1D CNN $\rightarrow$ BiLSTM” stack is our baseline [5] [8] [14].

C. Motion-Saliency Visualization (analysis-only)

For illustrative purposes, we visualize per-frame change using normalized frame differences:

$$\Delta x_t = \begin{cases} 0, & t = 1, \\ \text{norm}(x_t) - \text{norm}(x_{t-1}), & t \geq 2, \end{cases} \quad (4)$$

where $\text{norm}(\cdot)$ is per-frame channel-wise standardization. We use $|\Delta x_t|$ as an illustrative motion-saliency visualization in CSLR videos (Fig. 1); it does not require additional labels and is not used as an additional supervised signal in our training objective. Classic motion-energy ideas using frame differences have been used widely in vision [17].

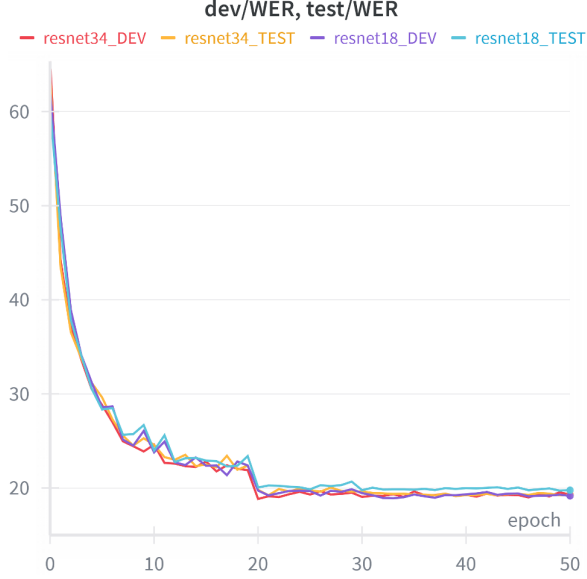


Fig. 3. DEV/TEST WER curves for the two backbone-temporal baselines, indicating stable convergence under the training recipe.

D. CTC Classifier and Self-Distillation (KD)

A shared linear classifier produces per-time-step logits for a vocabulary size V (including the blank). From the final sequence representation Z (after BiLSTM), we obtain teacher logits $\ell_t = W_c z_t + b_c \in \mathbb{R}^V$. We also attach an auxiliary CTC head to an intermediate representation (e.g., after the temporal CNN) to obtain student logits $\ell_t^{\text{aux}} = W_c u_t + b_c \in \mathbb{R}^V$, where the classifier parameters (W_c, b_c) are shared across the final and auxiliary heads. Let $y_{1:L}$ be the ground truth gloss sequence.

The sequence loss is CTC [9]:

$$\mathcal{L}_{\text{CTC}} = \text{CTC}(\log \text{softmax}(\ell_{1:T}), y_{1:L}). \quad (9)$$

To stabilize early training, we optionally attach an auxiliary CTC head to an intermediate representation (e.g., after the temporal CNN) to produce logits ℓ_t^{aux} . We distill ℓ_t (teacher) into ℓ_t^{aux} (student) with temperature τ [8] [15], inspired by self-distillation in CSLR [5]:

$$\mathcal{L}_{\text{KD}} = \tau^2 \text{KL}(\text{softmax}(\frac{\ell}{\tau}) \parallel \text{softmax}(\frac{\ell^{\text{aux}}}{\tau})). \quad (10)$$

The full loss is

$$\mathcal{L} = \mathcal{L}_{\text{CTC}} + \lambda_{\text{KD}} \mathcal{L}_{\text{KD}}. \quad (13)$$

We train end-to-end with Adam and the same schedule as the baseline.

IV. EXPERIMENTS

A. Dataset and Metric

We follow standard CSLR practice and evaluate on PHOENIX-2014 (German Sign Language weather/news) [18].

TABLE I
BASELINE RESULTS ON PHOENIX-2014 (WER WITH DELETION/INSERTION BREAKDOWN).

Method	Backbone	Dev WER	del/ins Dev	Test WER	del/ins Test
B0	ResNet-34	18.8	6.5/2.3	19.8	6.3/2.2
B1	ResNet-18	19.1	6.7/2.3	20.0	6.6/2.1
VAC [16]	ResNet-18	21.2	7.9/2.5	22.3	8.4/2.6
SignGraph [8]	ResNet-18	18.2	4.9/2.0	19.1	4.9/2.0
CorrNet [5]	ResNet-18	18.8	5.6/2.8	19.4	5.7/2.3
OLMD [7]	ResNet-34	17.1	4.8/2.4	17.2	4.7/2.3

The metric is Word Error Rate (WER), computed as $\text{WER} = (\# \text{sub} + \# \text{ins} + \# \text{del}) / \# \text{ref}$ (lower is better), where sub, ins, del denote substitution, insertion, and deletion, and ref is the number of reference glosses. In this paper, WER, del, and ins are reported as percentages unless stated otherwise.

B. Implementation Details

We use Adam with base learning rate 1×10^{-4} and weight decay 1×10^{-4} , and follow a cosine/step scheduler. The batch size is 4. Frames are resized to 224×224 with standard random crop/flip and mild temporal rescaling. Training is end-to-end with CTC. We optionally use self-distillation (KD) with temperature $\tau = 8$, where the *teacher* logits are from the final CTC head (after BiLSTM) and the *student* logits are from the auxiliary CTC head attached to the intermediate representation (after the temporal CNN). The final and auxiliary heads share the same classifier parameters. All training is performed with bfloat16 precision. For decoding, beam search on CTC outputs is used.

C. Baselines

Our primary baselines follow the common CSLR recipe:

- **B0 (ResNet-34):** ResNet-34 $\rightarrow$ TCN $\rightarrow$ BiLSTM $\rightarrow$ CTC.
- **B1 (ResNet-18):** ResNet-18 $\rightarrow$ TCN $\rightarrow$ BiLSTM $\rightarrow$ CTC.

For context, we also list representative published results (VAC [16], SignGraph [8], CorrNet [5], OLMD [7]) as reported in the corresponding papers. We also plot the training dynamics in Fig. 3, showing the DEV/TEST WER trajectories for both backbones under the same recipe.

D. Results on PHOENIX-2014

Table I summarizes baseline performance on PHOENIX-2014. Our ResNet-18/34 backbone+temporal baselines achieve test WER around $\sim 20\%$, which is consistent with the typical performance band for single-stream CTC-based systems. In addition to WER, we report deletion and insertion components to characterize how errors distribute under this standard recipe. Rows VAC/SignGraph/CorrNet/OLMD are taken from the corresponding papers and included for reference.

E. Motion-Saliency Visualization

We include a normalized frame-difference visualization (Fig. 1) as an illustrative motion-saliency view of signing videos. This figure is provided to support intuition about where pixel-level motion is strong and is not used as an additional supervision signal in training.

V. LIMITATIONS AND FUTURE WORK

A. Limitations

- This paper focuses on establishing and documenting strong, reproducible CTC baselines and error decompositions on PHOENIX-2014; broader benchmarking across additional datasets and settings is outside the current scope.
- Our diagnostic visualization uses simple normalized frame differences, which capture high-motion regions but can also reflect camera/illumination changes and may miss subtle articulations with low pixel variation.
- Our analysis is based on dataset-level error decomposition (WER with deletion/insertion components). We do not use frame-level boundary annotations or explicit boundary-detection metrics, so our diagnostics remain qualitative rather than a quantified boundary benchmark.

B. Future Work

- Extend evaluation to test cross-dataset (e.g., CSL-Daily) generalization and confirm whether a similar error pattern persists under domain shift.
- Develop richer but still lightweight diagnostics (e.g., temporal gradients, local phase-based descriptors, etc.) and measure their correlation with error occurrences to characterize “boundary-difficult” segments.
- Investigate training and decoding strategies that explicitly target deletion/insertion behavior (e.g., alignment-aware regularization, decoding constraints, or language-model integration) while keeping the baseline backbone unchanged.

VI. CONCLUSION

We presented a boundary-focused empirical view of CTC-based CSLR on PHOENIX-2014. By reporting reproducible backbone+temporal baselines together with deletion/insertion breakdowns and training-stability curves, we highlight persistent alignment fragility around ambiguous transitions and coarticulation. We also provided a lightweight motion-saliency visualization (normalized frame differences) as an illustrative diagnostic to support discussion and visual inspection. We hope these baselines and diagnostics help the community more consistently characterize where CTC pipelines fail and motivate future boundary-aware objectives and decoding strategies for improving CSLR robustness.

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